Name John Guast

Date 10/30/23 Lab Instructor/Section

Data sheet

Parts I and III (Show calculations in margin or below label)

1. Initial volume of graduated cylinder 58.0 mL
2. Final volume of graduated cylinder 84.0 mL mL
3. Volume of H_2 gas evolved 84.0 mL − 58.0 mL = 26.0 mL
4. Volume of H_2 gas evolved 0.0260 L
5. Temperature of water (and gas) 22.0 °C
6. Temperature of water (and gas) 295.15 K
7. Atmospheric pressure 756 mm Hg
8. Vapor Pressure of water at experiment temperature (table) 19.8 mm Hg
9. Pressure of H_2 736.2 mm Hg
10. Pressure of H_2 0.967 atm
11. Moles of H_2 from ideal gas law $1.04E^{-3}$ mol
12. Moles of Mg from reaction stoichiometry $1.04E^{-3}$ mol
13. Mass of Mg metal reacted $1.04E^{-3}\ mol \cdot \frac{24.31\ g}{1\ mol} = 0.0252\ g$

$1.04E^{-3} \frac{1\ Mg}{1\ H_2} = 1.04E^{-3}\ mol$

$PV = nRT$

$\frac{0.967\ atm \cdot 0.0260\ L}{0.0821 \cdot 295.15\ K} = n = 1.04E^{-3}$

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